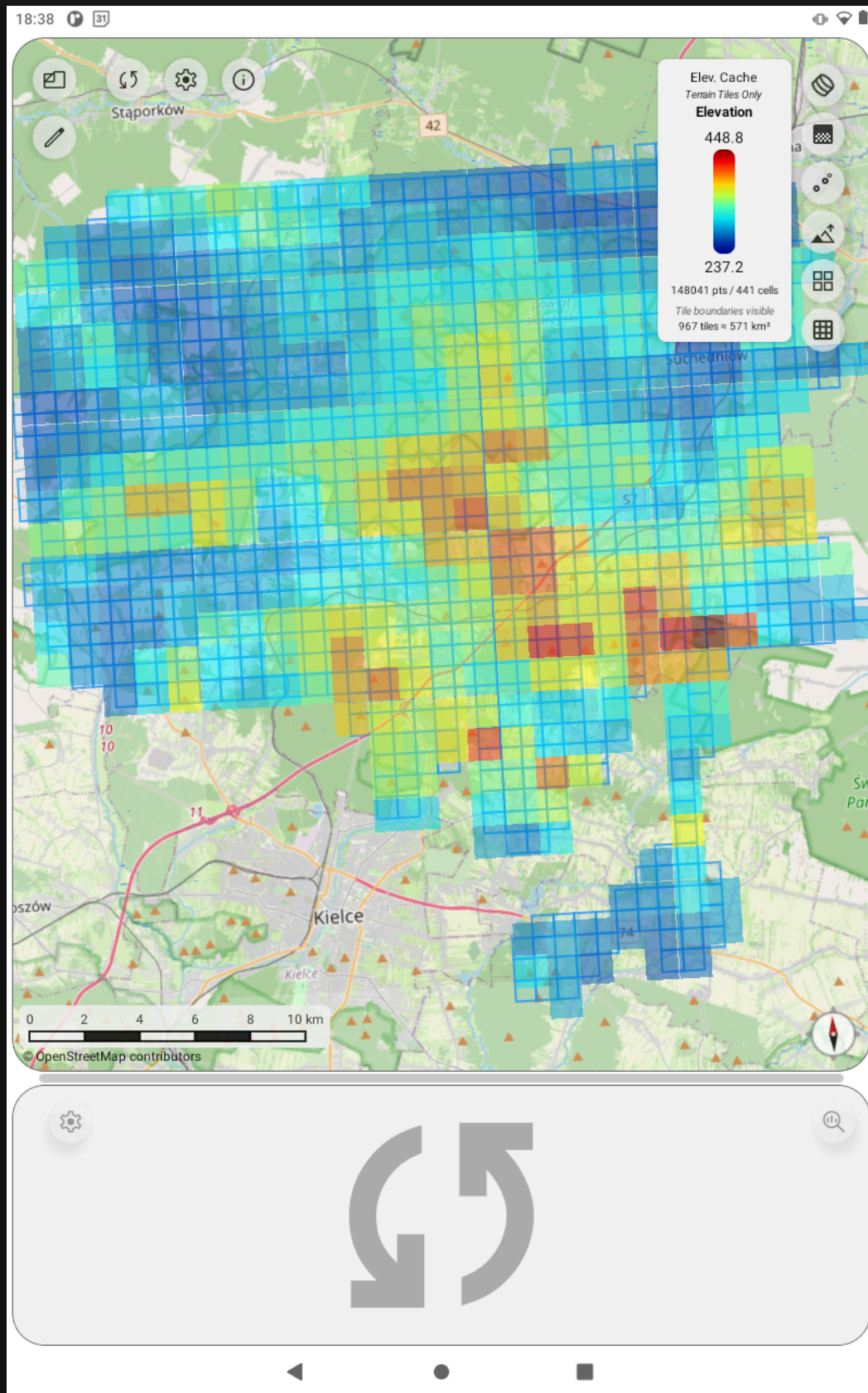




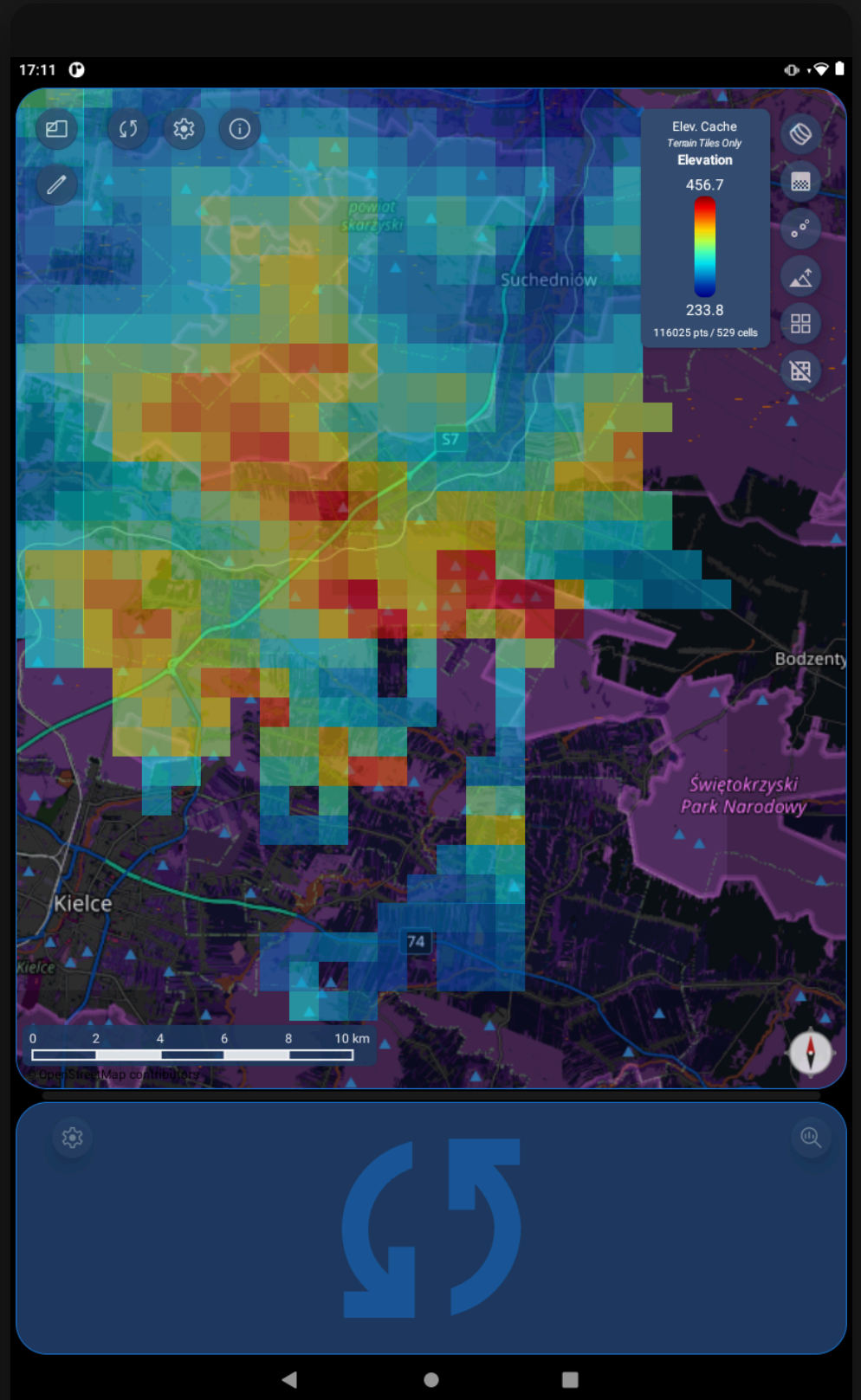
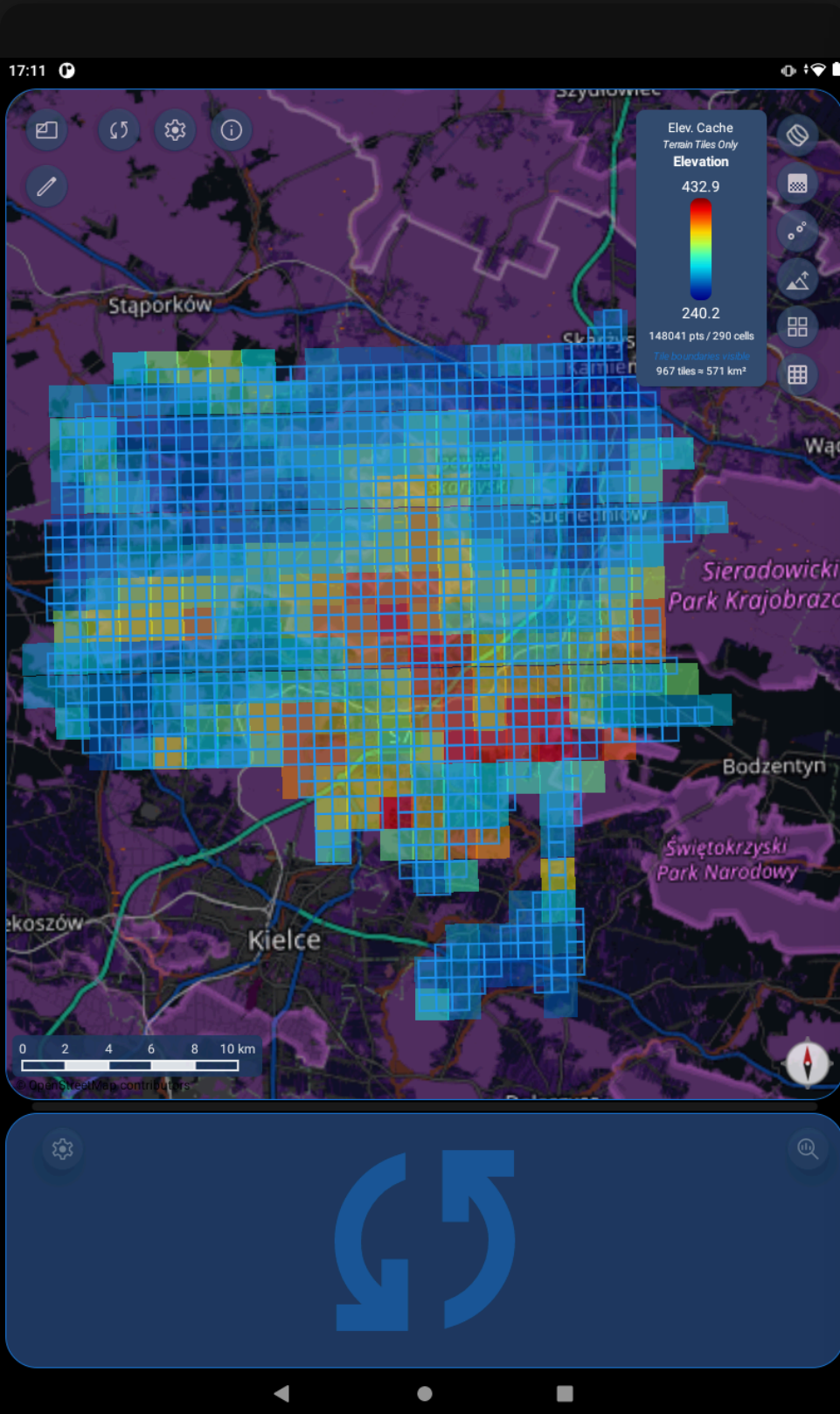
v2.16.0

Offline Elevation — Terrain Tile Provider

Fetch once, use **offline forever**.

AWS Terrain Tiles decoded on-device. ~30m source resolution.

Each tile covers $\sim 9.5 \text{ km}^2$ · ~50 KB cached locally.

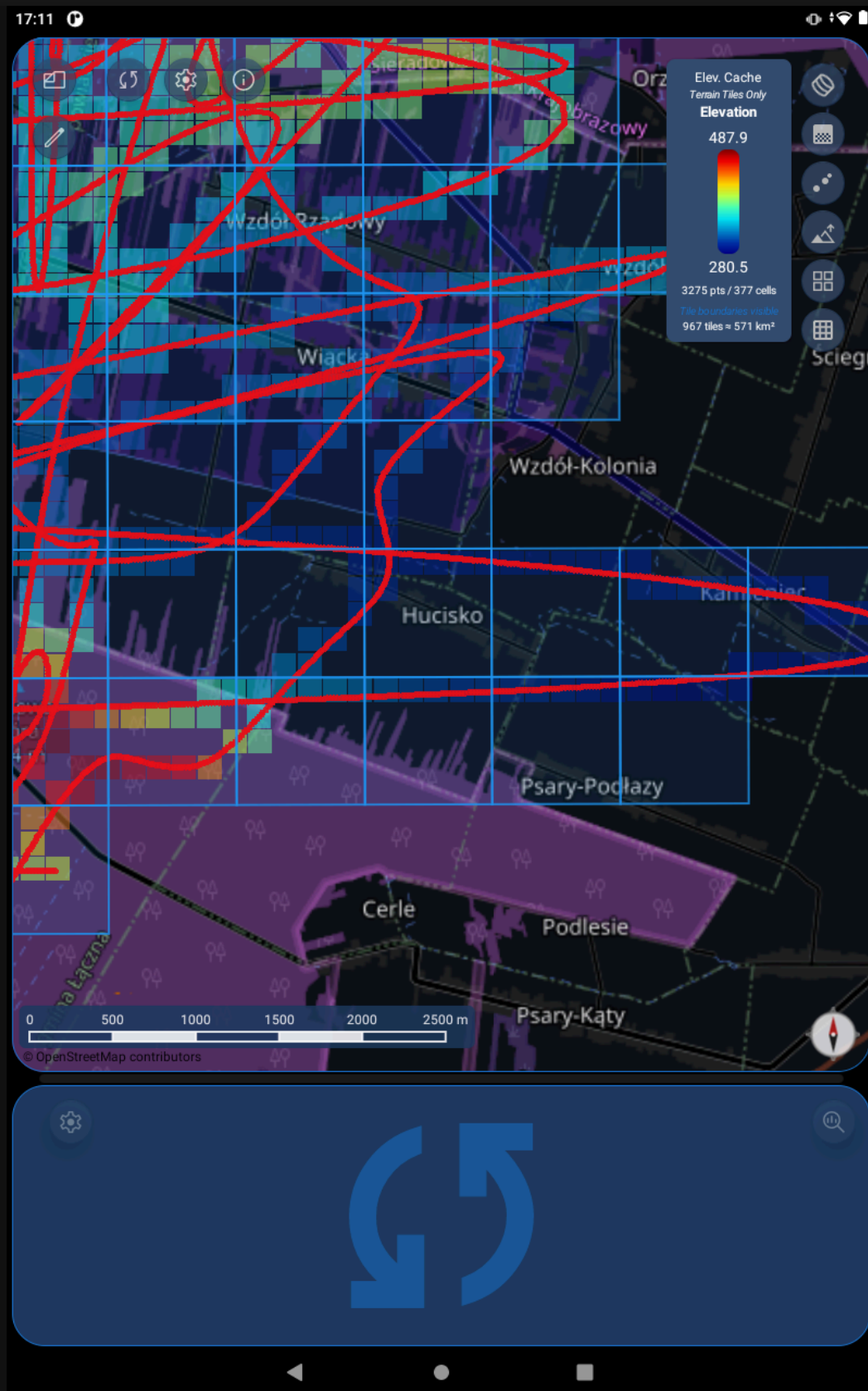


Tile Boundary Overlay — See Your Coverage

Toggle tile grid to visualize cached terrain tiles.

Blue rectangles = cached. Tile count + area in legend.

Left: boundaries ON · Right: boundaries OFF.



Trail-Level Detail — Tiles + Heatmap

Elevation points decoded from **RGB pixel channels**.

Three-layer cache: in-memory LRU → Room DB → AWS S3.

Sub-millisecond lookups on cached tiles.



Built for Speed

450× FASTER CACHE WRITES

Smart spatial indexing replaces brute-force lookups.

1,561 elevation points saved in **< 1 second** instead of 38s.

3-LAYER CACHE — INSTANT REPEAT LOOKUPS

Memory → local database → cloud, automatically.

Previously visited areas load in **sub-millisecond** time.

ONE TILE DECODE SERVES HUNDREDS OF POINTS

Nearby points share the same tile image (~9.5 km²).

No repeated downloads. No redundant processing.

SMOOTH MAP RENDERING AT ANY SCALE

Overlay data updates without blocking the map.

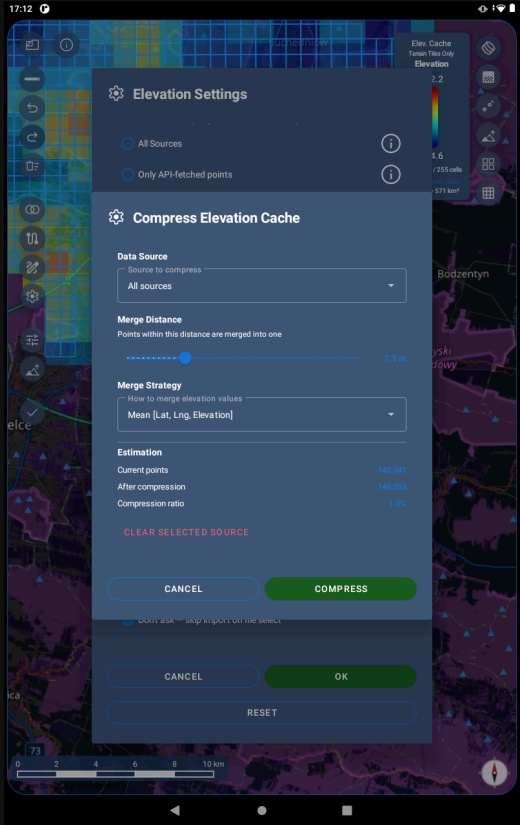
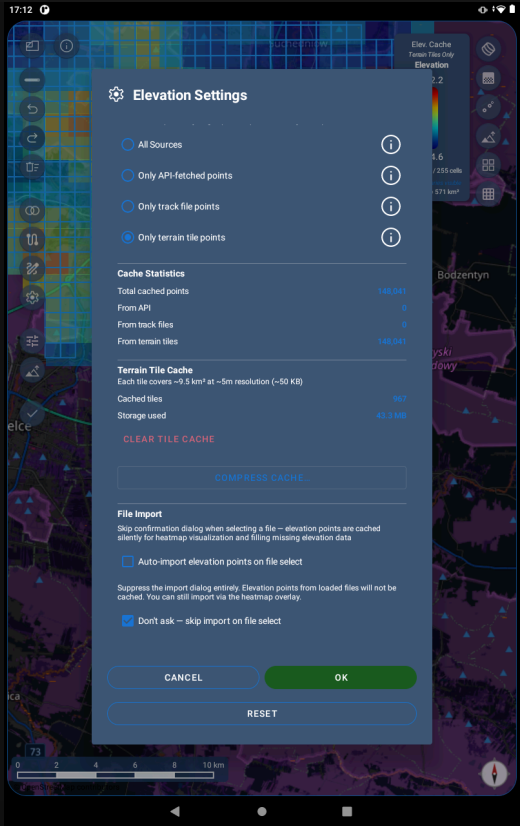
10,000+ point blind spot regions render without lag or crashes.



Background Tile Prefetch

WorkManager-based download · survives app death.
Real-time progress. Bounded concurrency (4 tiles).
Coverage grows automatically as you explore.

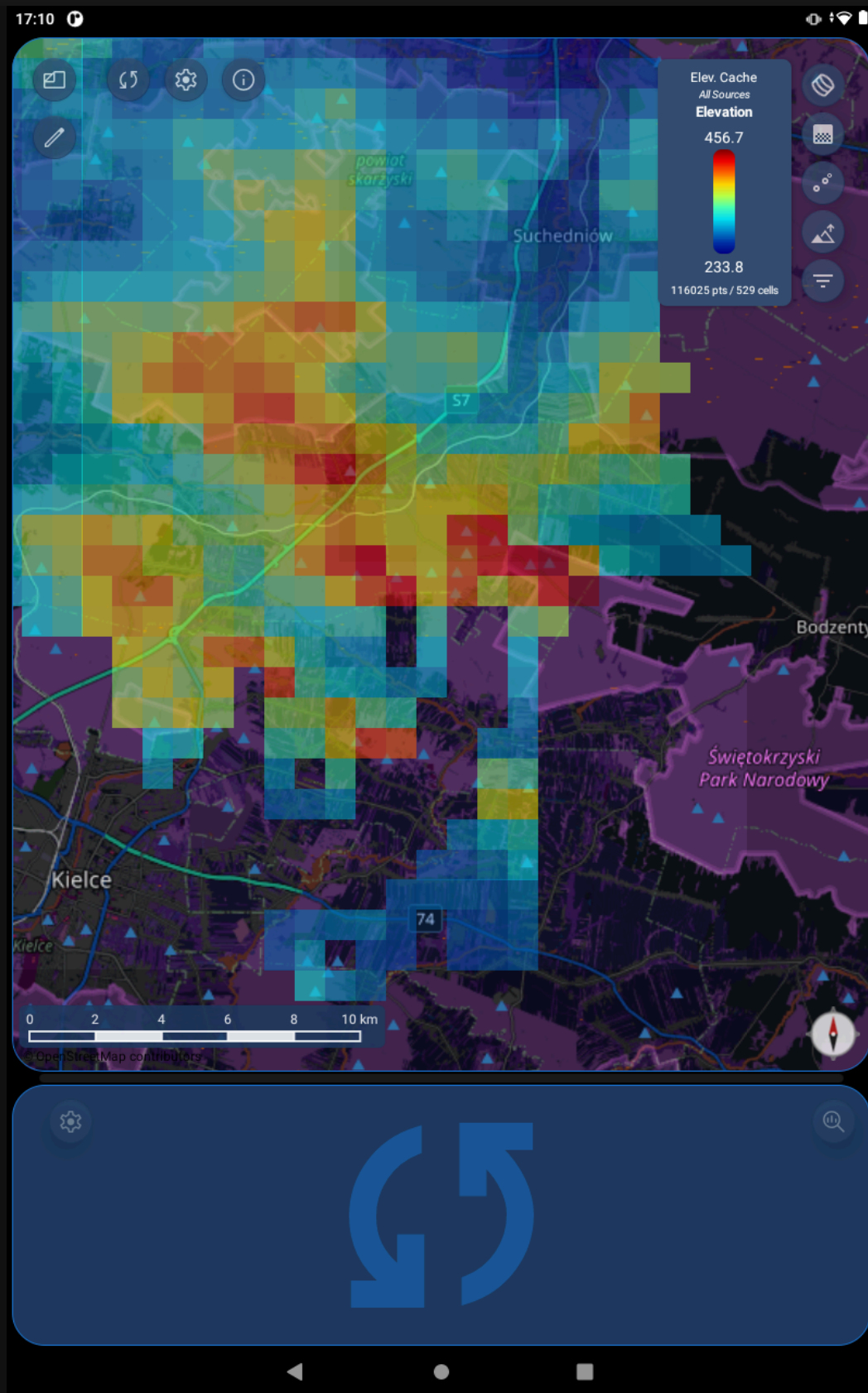
Red polygons = areas where elevation failed.
Info dialog explains cause: offline, tile voids, uncached.
 Spatial grid clustering · convex hull rendering.



Full Elevation Pipeline Control

Data Source · Fetch Mode · Cache Origin Filter

5 strategies · cache stats · tile management · compression.

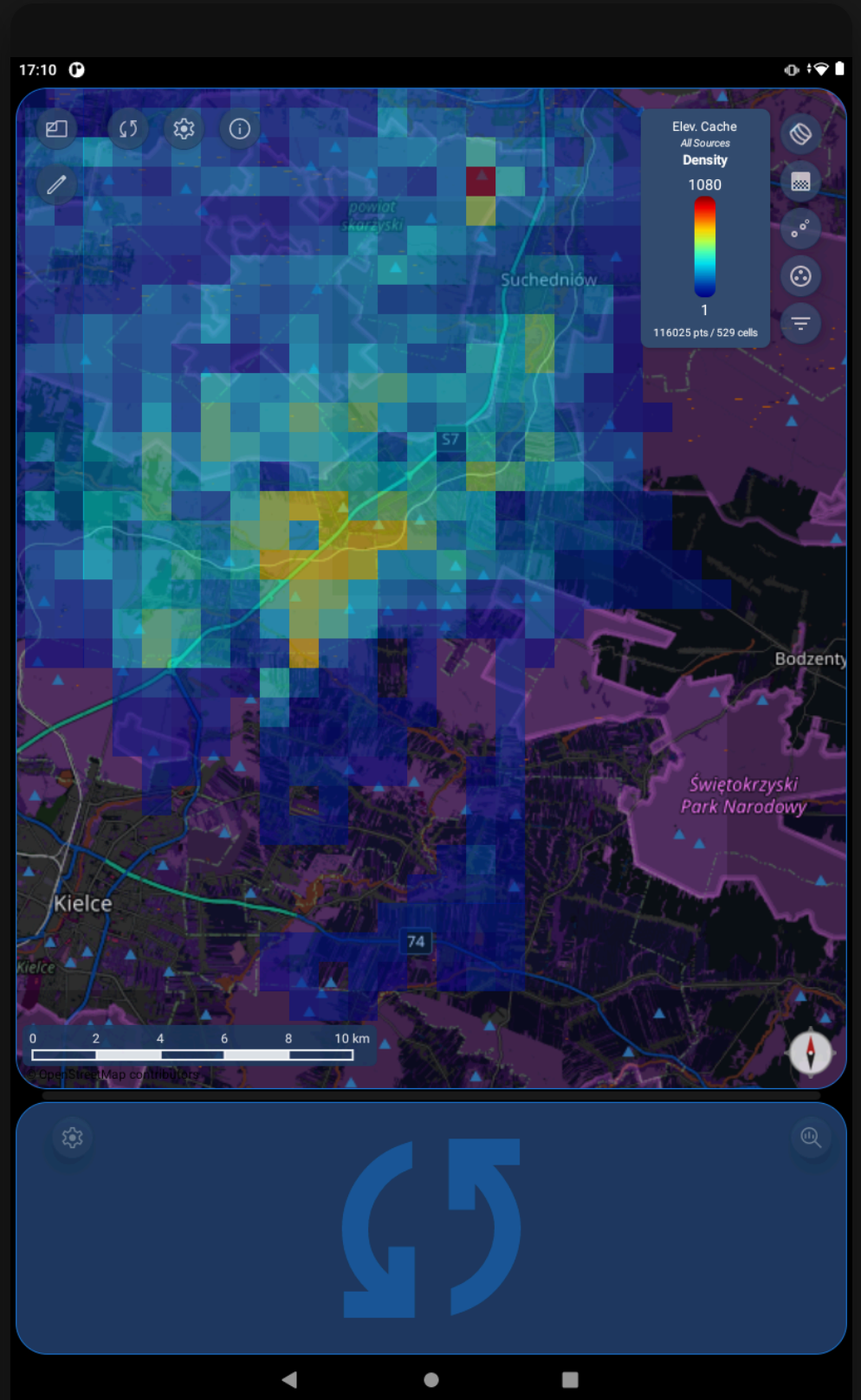
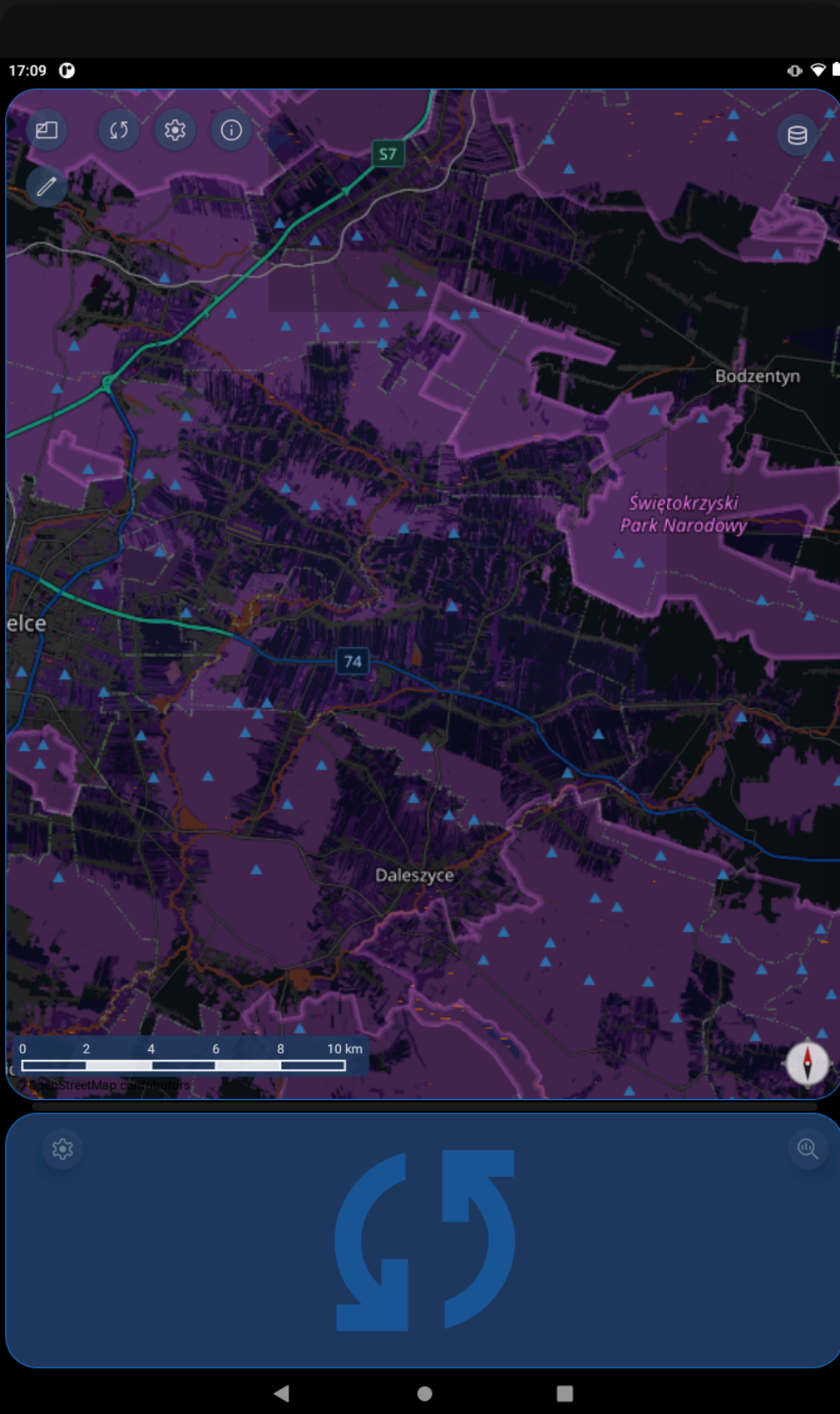


4-State Source Filter — API / Track / Tile / All

Elevation cache tracks **3 data origins**.

Filter by source in heatmap overlay and cache-only mode.

Cycle with one tap. New tile-source icons.

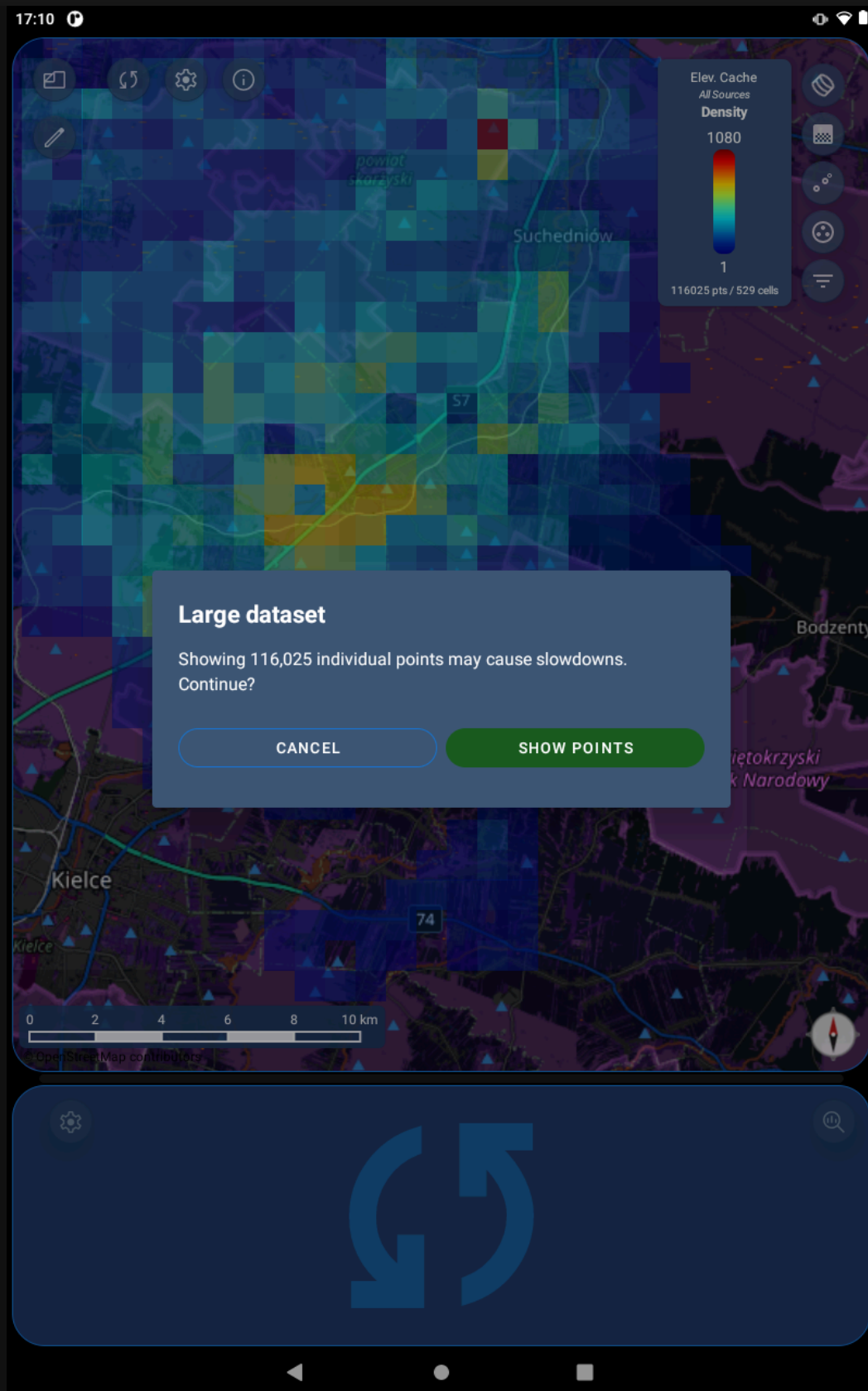


Refreshed Overlay UI — Material Symbols

Database · Gradient · Group Work icons.

Programmatic FABs — **no XML per layout variant.**

Adaptive legend. Consistent 48dp touch targets.

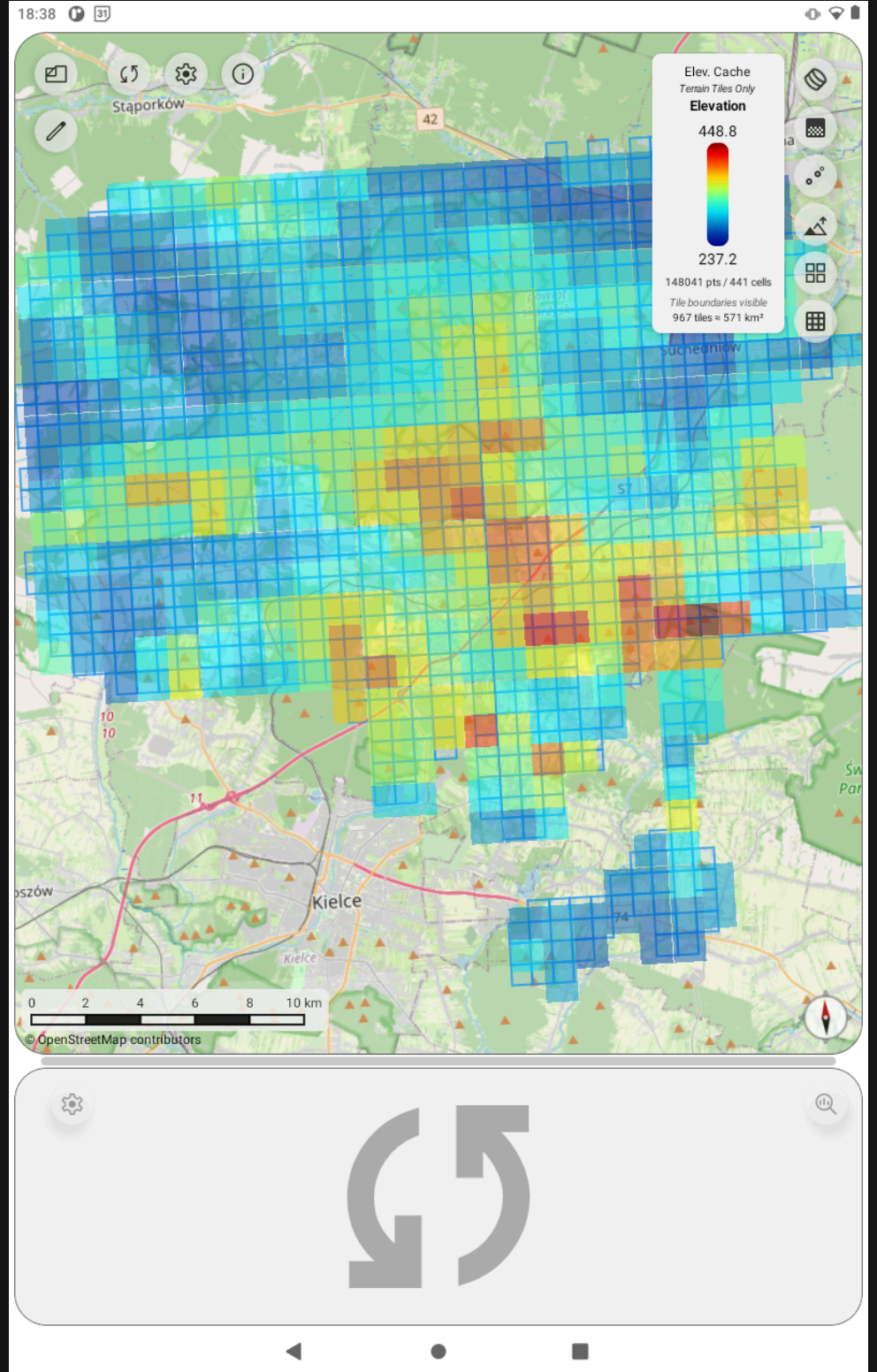


Graceful Offline Degradation

Strategies never crash offline — **partial results + NaN**.

Tile-resolved data preserved when API fallback fails.

"Don't ask again" for import · skip estimation.



v2.16.0 — Offline-First Elevation

Terrain Tiles · Blind Spot Detection · Cache Compression
5 fetch strategies · 3 data sources · background prefetch

Open to B2B contract or long-term cooperation

Connect or DM → [linkedin.com/in/stanczuk](https://www.linkedin.com/in/stanczuk)